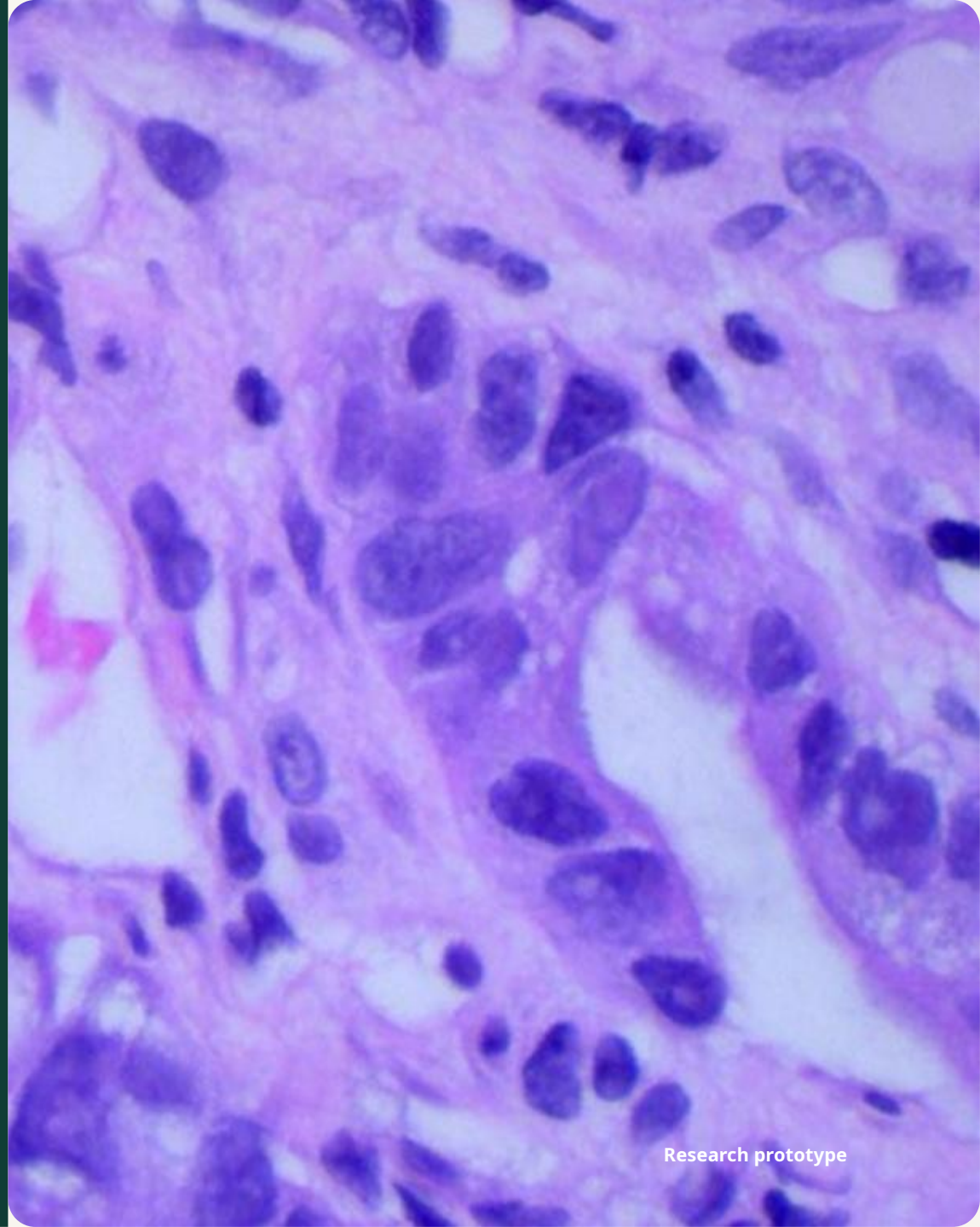
HistoPrompt

Model-aware prompting for pathology review

2026 Ashby Prize Hackathon

Built on PathGen, four NCSA pathology models, Delta, and Lumen



Five disciplines, one pathology workflow

Our team combines quantitative science, software, biomedical engineering, and domain-oriented evaluation.

ASTRONOMY • BIOENGINEERING • COMPUTING • PHYSICS • MATHEMATICS

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COMPUTING + BIOENGINEERING

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MATHEMATICS

Together, we built and evaluated a single clinician-facing system.

What Lumen changes for each candidate

The baseline names the diagnosis. The enhanced version adds a short, visually discriminative morphology phrase.

BASELINE PROMPT

a histopathology image showing colon adenocarcinoma

→

LUMEN + RETRIEVED EVIDENCE

colon adenocarcinoma with irregular glandular structures, cribriform patterns, desmoplastic response

1 Read case context

The report and differential define what needs to be distinguished.

2 Retrieve local evidence

The system supplies relevant morphology and diagnostic context.

3 Generate eight prompts

Every candidate keeps its exact label and receives a concise feature phrase.

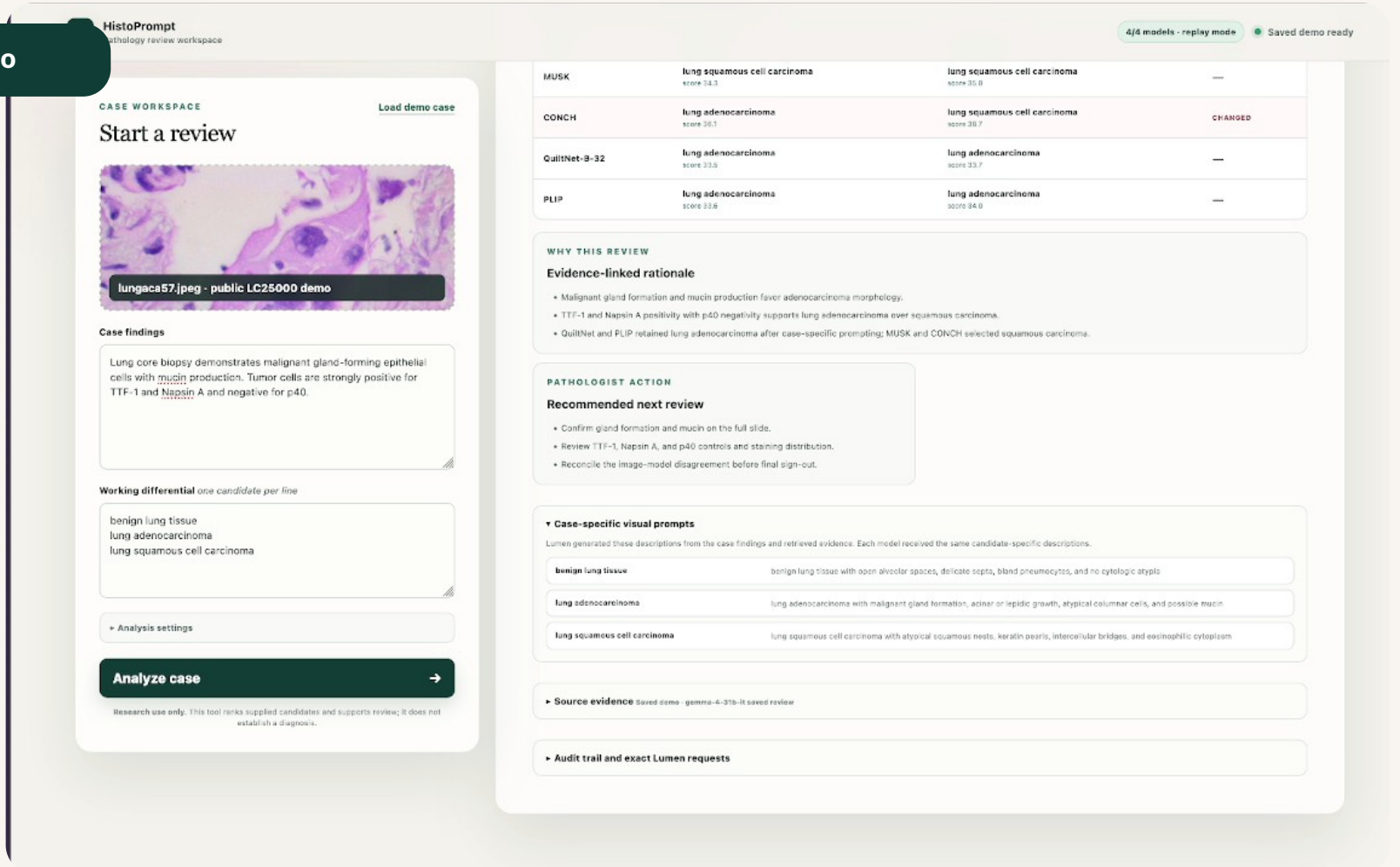
HistoPrompt measures this transformation and learns which model–organ pairings benefit.

Source: saved case record 4cff975b-adbe-4ecf-9be5-365c715e2305_53792_26368.json

The prototype is organized around a single review

Inputs stay visible while model results, evidence, and next actions are reviewed.

Public saved demo



Source: saved replay in data/demo_replay.json; demo video

Built on the full hackathon stack

We combined PathGen, RAG, Lumen, four pathology models, and Delta in one working product.

01

PathGen

2,141 pathology cases
8 TCGA cancer projects

02

NCSA models

MUSK · CONCH
QuiltNet · PLIP

03

Delta

Hosted inference
parallel model
evaluation

04

RAG + Lumen

Retrieved pathology
evidence
case-specific prompts

17,128

candidate-specific prompts

17,128

baseline / enhanced decisions

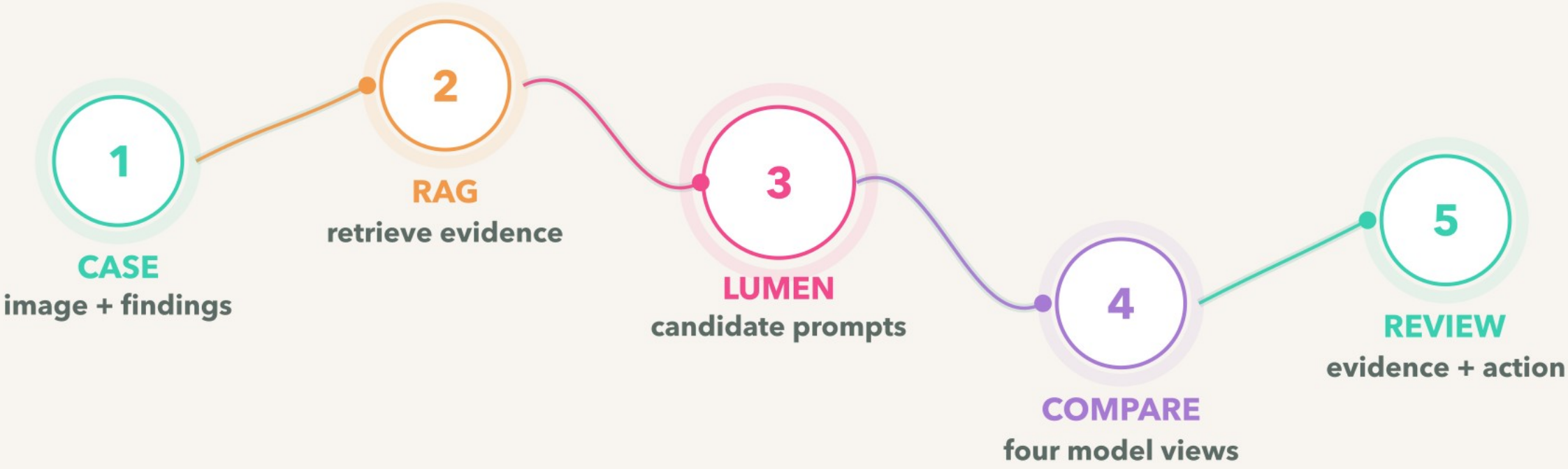
2 hours

end-to-end benchmark

Source: repository implementation; CCIL hackathon resources

One clinical case. Five traceable stages.

The same traceable chain powers the app, saved demo, and 2,141-case benchmark



Every score keeps its evidence and review path visible.

Ground-truth labels were never sent to Lumen

Two days produced a complete, testable system

The project moved beyond a UI mockup into live model integration, scripted evaluation, and portable delivery.

01	**Prototype**	Clinician case entry and review UI
02	**Integration**	MUSK, CONCH, QuiltNet, PLIP, and Lumen
03	**Evaluation**	Small screens, held-out tuning, and full PathGen workflow
04	**Reproduction**	Checksums, resumable runs, saved replay, and reports

30
commits

60
automated tests

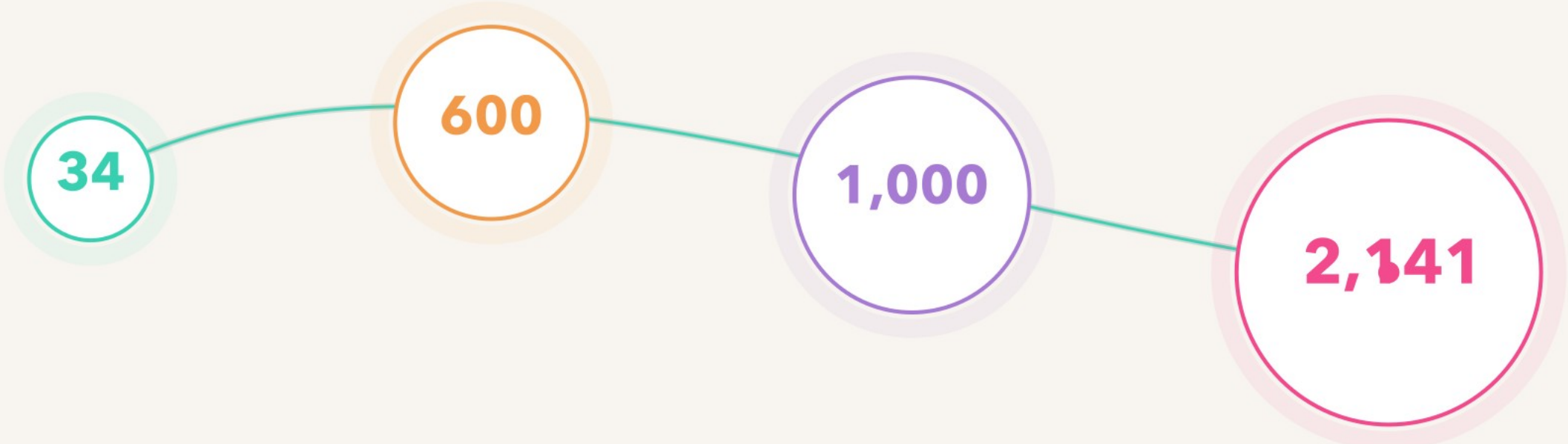
1
saved replay

Everything shown in this document is backed by saved code, data, or replayable artifacts in the private repository.

Source: repository history through f378abc

The final benchmark covered 2,141 cases

Model coverage reached four; cancer coverage reached eight



SCREEN
4 clinical groups

STRESS TEST
3 balanced classes

MODEL SWEEP
4 hosted models

FULL WORKFLOW
8 TCGA cancers

CASES
34 → 2,141

MODELS
1 → 4

COVERAGE
4 groups → 8 cancers

The full run identified six high-gain model–organ pairings.

Six cancer contexts. Three model routes.

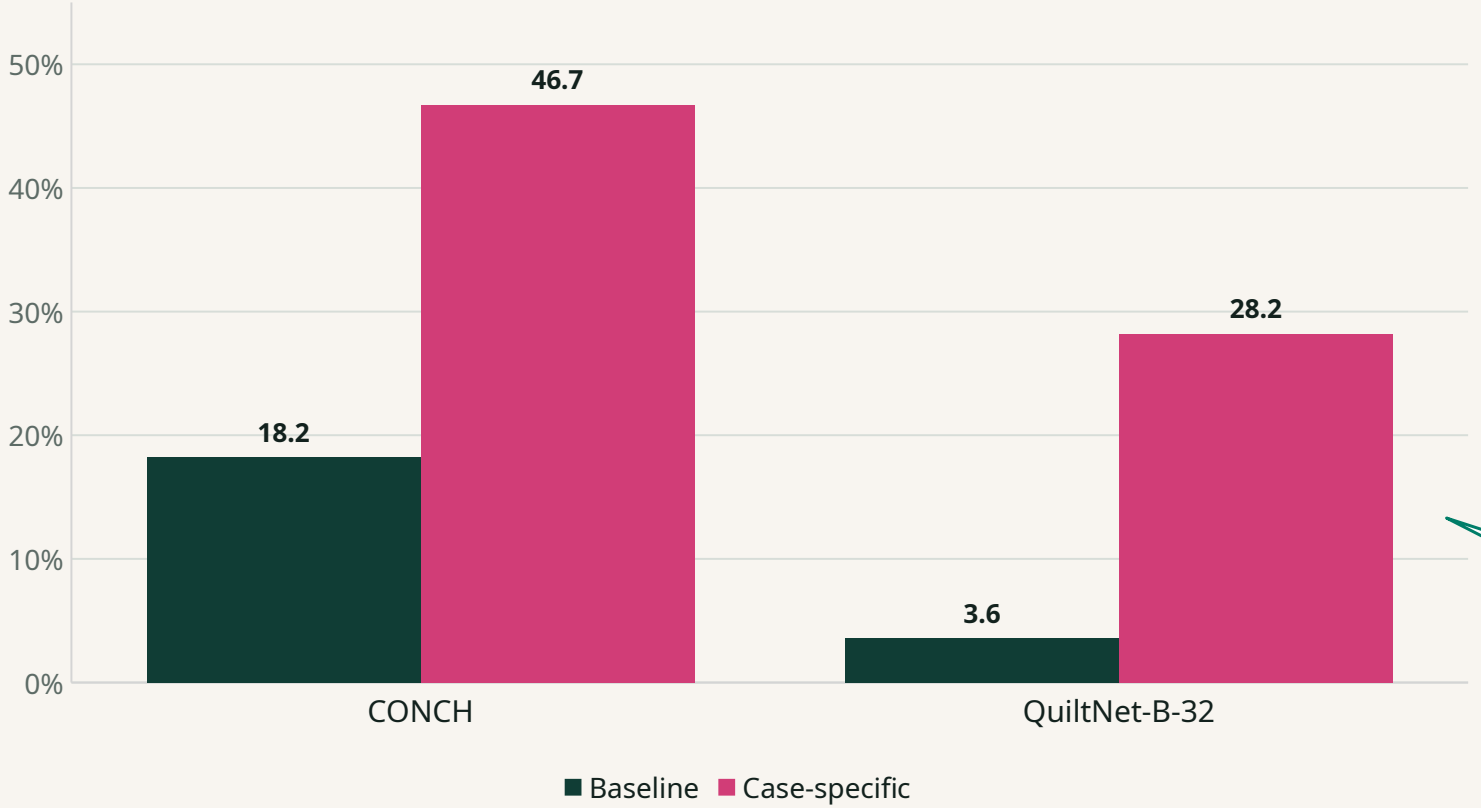
Each cancer context connects to the model with the largest observed gain



2,141 PathGen cases · eight TCGA projects · four hosted pathology models

Colon was the clearest multi-model success

Two independent pathology models improved substantially on the same 422-case PathGen cohort.



+28.4

points · CONCH

+24.6

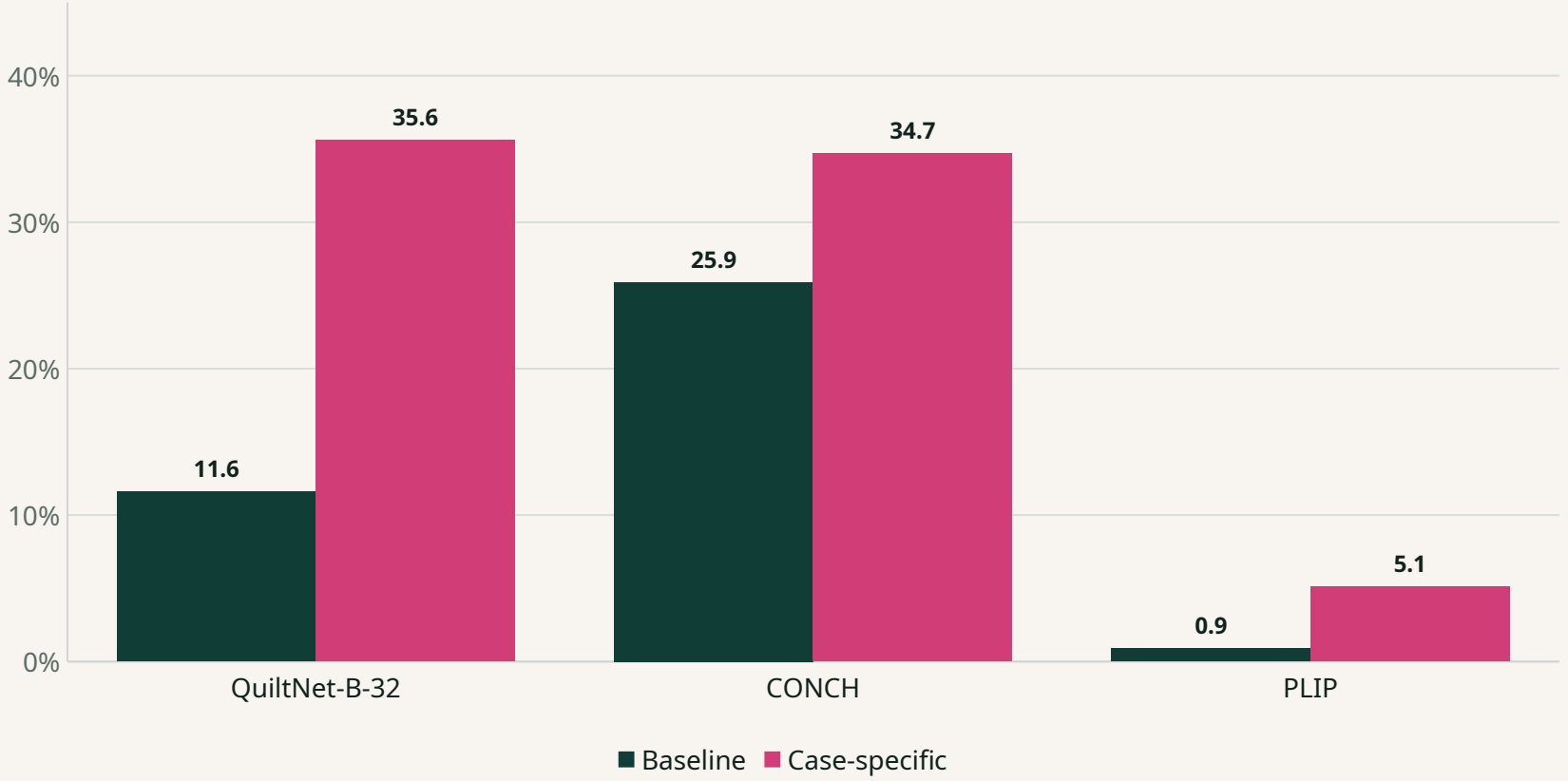
points · QuiltNet

Both models improve on the same 422-case cohort.

Source: TCGA-COAD subset, pathgen-clinical-workflow results

Breast showed gains across three models

QuiltNet delivered the largest improvement, with CONCH and PLIP moving in the same direction.



+24.1

QuiltNet points

+8.8

CONCH points

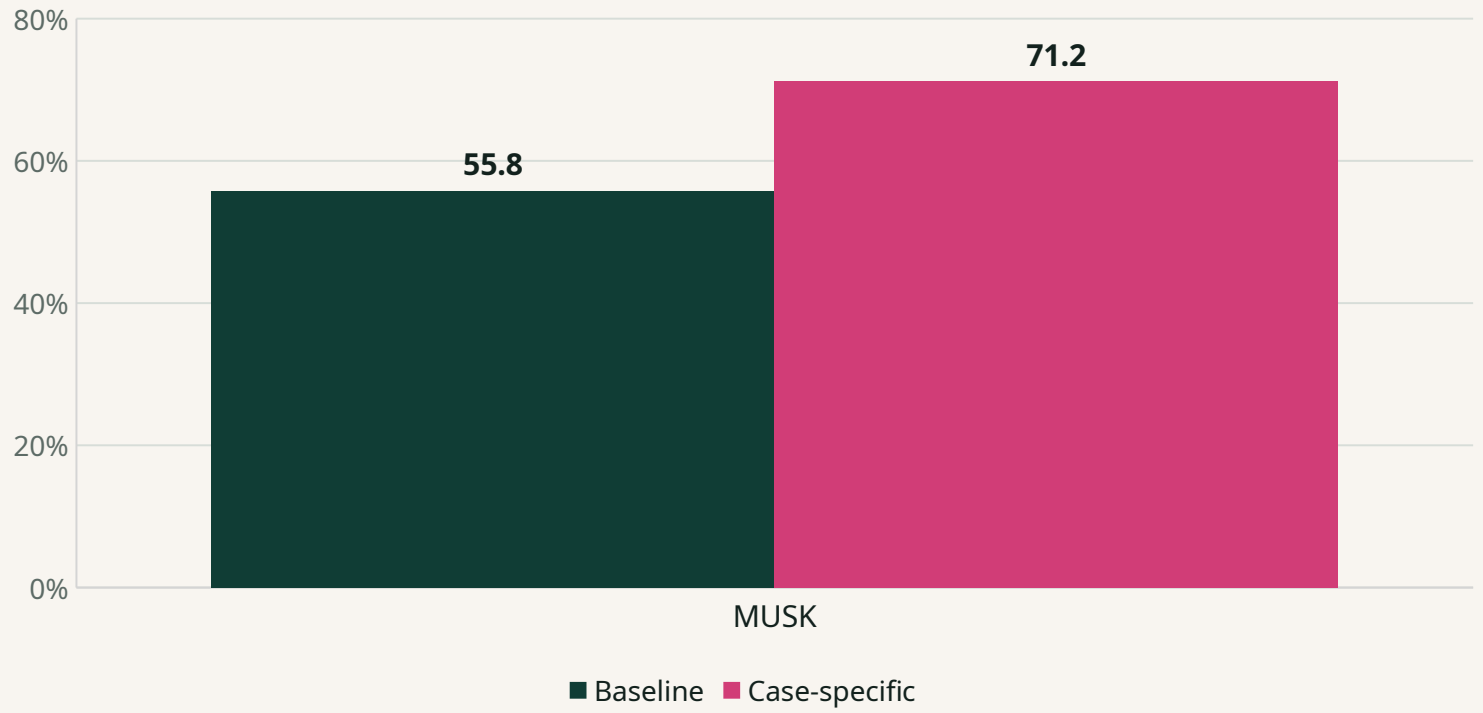
+4.2

PLIP points

Source: TCGA-BRCA subset, pathgen-clinical-workflow results

MUSK reached 71.2% on rectal adenocarcinoma

Case-specific prompting added 24 correct classifications on the 156-case cohort.



+15.4

percentage points

87 → 111

correct cases

Source: TCGA-READ subset, pathgen-clinical-workflow results

The routing pattern repeated in kidney and lung squamous

Kidney selected MUSK; lung squamous selected QuiltNet.

Kidney clear cell

MUSK

20.3% → **31.1%**

+10.7 points

89 → 136 correct

Lung squamous

QuiltNet

2.3% → **16.3%**

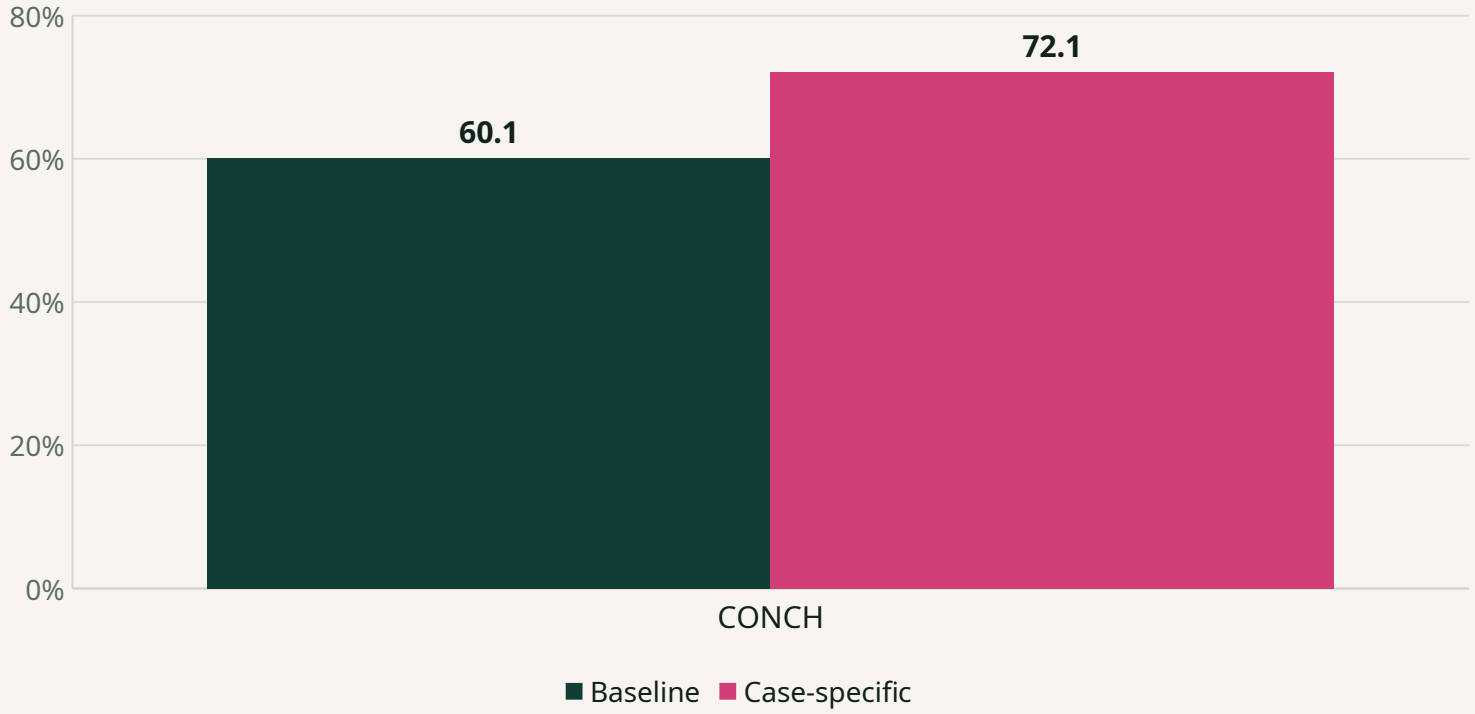
+14.1 points

6 → 43 correct

Source: TCGA-KIRC and TCGA-LUSC subsets

CONCH crossed 72% on prostate adenocarcinoma

The case-specific prompt improved accuracy by 12 points across 208 PathGen cases.



+12.0

percentage points

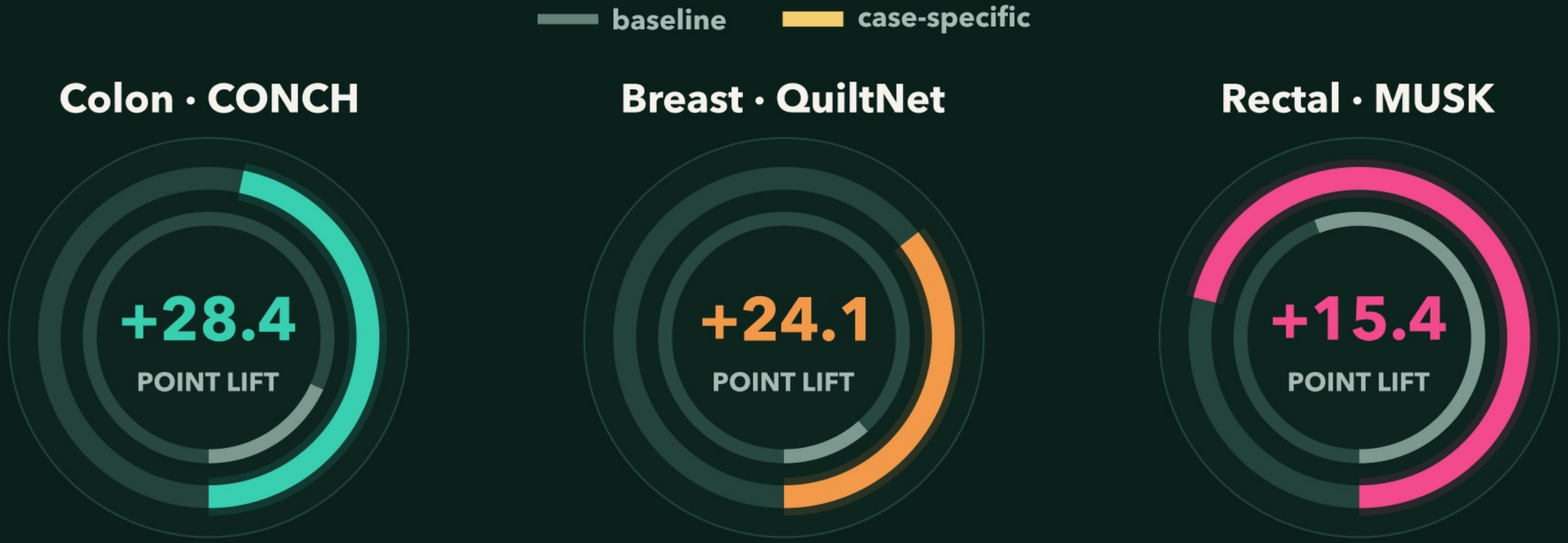
125 → 150

correct cases

Source: TCGA-PRAD subset, pathgen-clinical-workflow results

Three pairings delivered the largest gains

Outer ring: case-specific accuracy · inner ring: baseline accuracy



Colon → CONCH · Breast → QuiltNet · Rectal → MUSK

Saved 2,141-case clinician-workflow benchmark

Independent tuning added 4.4 points for two models

The search used a development set, confirmation set, and frozen 450-image held-out lung test for each model.



+20

MUSK correct cases

+20

PLIP correct cases

Each model received its own prompt search.

Source: benchmarks/lc25000-tuning results

One tuned template added 66 correct MUSK predictions

Held-out evaluation covered lung, colon, CRC9 tissue classes, and PatchCamelyon metastasis.

Four tasks

- LC25000 lung
- LC25000 colon
- CRC9 tissue classes
- PatchCamelyon metastasis

MUSK held-out total

1,444 → **1,510**

baseline correct | tuned correct

+66 correct

CRC9: 319 → 364 correct

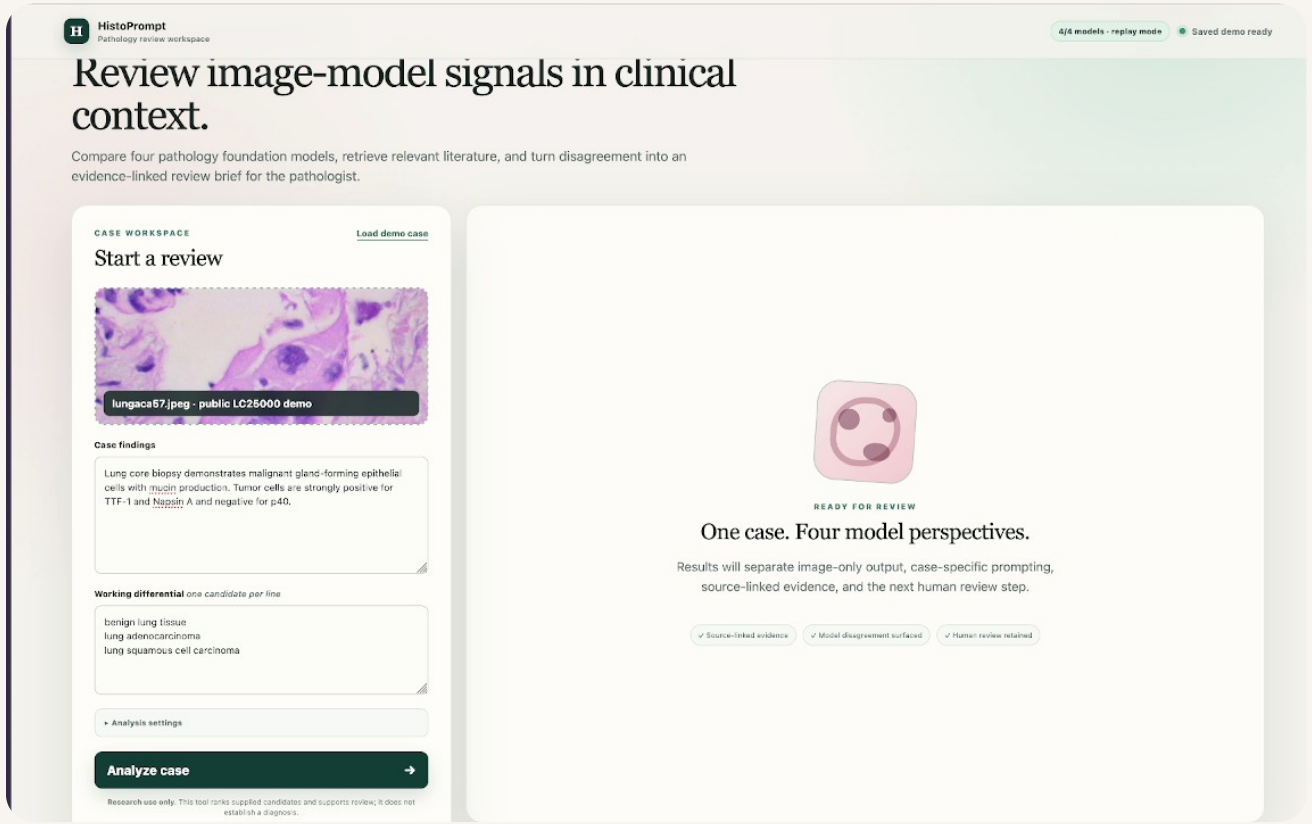
Shared structure + task-specific language

Shared structure and task-specific phrases produced the 66-case gain.

Source: origin/validated-prompt-tuning at 5ae7423

One screen supports a fast clinical review

Inputs, model rankings, evidence, and next actions stay visible.



01 **Compare**

Four model perspectives on the same differential

02 **Trace**

Prompts, retrieved evidence, and source passages

03 **Act**

A concise rationale and next review step

Source: implemented UI and saved demo

The full benchmark is reproducible

A reusable script preserves the inputs, outputs, checksums, and workflow state.

2,141

case records

Exact report, prompts, model outputs, and review

SHA-256

checksums

Images, reports, corpus, and workflow inputs

1

fingerprint

Models, candidates, retrieval settings, and source

resume

safe reruns

Completed matching cases are skipped

report-only

instant summaries

No new endpoint calls required

60

automated tests

Pipeline, tuning, backends, replay, and setup

One command runs the app. One command rebuilds the benchmark report.

Source: scripts, manifests, case records, workflow fingerprints, tests

HistoPrompt finds where prompting works

and puts those gains into a usable workflow.

+28.4

CONCH · colon

+24.1

QuiltNet · breast

+15.4

MUSK · rectal

Built with

PathGen
Delta
MUSK
CONCH
QuiltNet
PLIP
Lumen

2,141 cases • 8 cancers • 4 models • 1 reproducible workflow

Next: prospective confirmation of the six model-aware routing targets.

github.com/osirais/histoprompt

Clinician prototype + saved demo + full benchmark